Yiran Yuan

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Research Interests

Statistical and AI methods for complex systems, with applications to scientific discovery networks and trustworthy biomedical AI; network diffusion, auditable evidence synthesis, uncertainty quantification, and calibrated evaluation.

Education

Southern University of Science and Technology, B.S. in Statistics — Sep. 2023 – Jul. 2027 (expected)

- GPA: 3.72/4.0; rank: 10/30
- **Quantitative foundations:** Mathematical Analysis I–II, Linear Algebra, Probability Theory, Operations Research and Optimization, University Physics II, Basic Physics Laboratory
- **Statistics:** Mathematical Statistics, Multivariate Statistical Analysis, Bayesian Statistics, Functional Data Analysis, Time Series Analysis, Financial Statistics, Statistical Computing and Software
- **Computing and applications:** Introduction to Data Science, Python Programming Fundamentals, Data Structures and Financial Applications, Network Science and Computing, Business Big Data Analytics, Finance

University of Pennsylvania, Exchange Student in Statistics — Aug. 2025 – Dec. 2025

- Coursework: Forecasting for Time Series, Introduction to R

Research Experience

Scientific Discovery Birthplaces and Citation Diffusion in Global Knowledge Networks — Apr. 2026 – Present

- Built discovery-level OpenAlex corpora with entity disambiguation and non-research filtering; a pilot audit achieved 86% topical precision.
- Detected discovery-specific burst years with Kleinberg's algorithm (79% aligned with known takeoff periods within ± 5 years) and ranked foundation papers using citation-neighborhood expansion with semantic and citation signals (84% top-10 coverage of validated canonical works).
- Found a heavy-tailed pattern in citing-city ranks ($R^2 = 0.88$ on a log–log rank-size fit), motivating ongoing analysis of structural mechanisms behind geographic knowledge diffusion; working paper in progress.

Auditable Evidence Synthesis Across HIV and Alzheimer's Disease — Mar. 2026 – Present

Remote Research Assistant, PI: Shana D. Stites, University of Pennsylvania

- Built an auditable PubMed pipeline comparing dense, BM25, and hybrid retrieval; linked extracted claims to source passages and added deterministic checks for incompatible coefficient definitions and scoring directions.
- Against a 27-study HIV reference set, retrieval recovered 25/27 studies; screening achieved 100% precision, 78% recall, 100% specificity, and identical fixed-input decisions across five runs (Fleiss' $\kappa = 1.00$).
- Constructed an evidence map covering 37 studies of people with HIV and 48 Alzheimer's disease studies; study-level labels remained grounded in citation evidence rather than unsupported pooled estimates.
- Developing a local-model, PHI-compliant workflow to code 104 de-identified clinical interview transcripts against a health-literacy codebook, beginning with a manually coded reference subset.

SymBreak: Subvisual Ischemic Change Detection from Non-Contrast CT — Jun. 2026 – Present

- Contributed to a counterfactual false-belief implantation framework that generates bilaterally symmetric cases from unilateral ischemic samples.
- Conducted patient-level Symmetry-Breaking Index analysis and evaluated robustness across five public NCCT datasets, four held-out acute ischemic stroke cohorts, and multiple segmentation architectures.
- Analyzed model responses and error patterns, including background imbalance in non-stroke cases and preserved symmetry breaking in unilateral ischemic stroke; manuscript under review at ICLR 2027.

Prevalence-Aware Calibration for Parkinson's Disease MRI — 2026

- Designed an exploratory leakage-safe 3D MRI framework combining bilateral representation learning, a healthy-control symmetry constraint, and class-conditional Bayesian posterior risk under an explicit target-prevalence prior.
- Pre-specified calibration, bootstrap uncertainty, site stability, and sensitivity analyses; no final clinical-performance result is claimed.

Industry Research Experience

Imaging Algorithm Evaluation Intern, OPPO Jun. 2026 – Oct. 2026

- Studied disagreement between objective video-quality measurements and expert severity ratings under latent scene structure and heterogeneous score distributions.
- Designed score-debiasing corrections and unsupervised scene clustering; trained CNN/Transformer models to predict expert judgments from video features.
- Improved objective–subjective consistency from 82.2% to 94.6% for downstream imaging-algorithm evaluation.

Manuscripts and Presentations

SymBreak: Breaking the Implanted False-Belief to Identify Subvisual Ischemic Changes 2027

- Conference manuscript under review at ICLR 2027.

Evidence-Oriented Mapping of Subjective–Objective Cognitive Dissociation in HIV and Alzheimer's Disease 2026

- Poster presented at the 2026 Penn CFAR Scientific Symposium, University of Pennsylvania.

Metacognitive Experience in People With HIV: A Systematic Review 2026

- Conference poster presented at AAIC Neuro Next 2026.

Additional Project

User Growth Strategy Optimization with Causal Inference Nov. 2025 – Jan. 2026

- Estimated user-level conditional average treatment effects with an X-Learner and optimized outreach allocation using quadratic programming under channel-budget and contact-frequency constraints.

Honors and Awards

Academic Excellence Scholarship, SUSTech 2023–2025
Outstanding Freshman Scholarship, SUSTech 2023
MCM/ICM, Successful Participant Award 2025
FLTRP Cup English Speaking/Writing Contest, Bronze Award 2024

Technical Skills

- **Statistics and networks:** Bayesian inference, calibration, uncertainty quantification, causal inference, uplift modeling, bibliometrics, citation networks, burst detection
- **Machine learning and evidence systems:** PyTorch, scikit-learn, BERT, Sentence-BERT, FAISS, dense and hybrid retrieval, CNNs, Transformers, 3D medical imaging
- **Programming:** Python, R, SQL, Git; Pandas, NumPy, SciPy, statsmodels
- **Languages:** TOEFL 5.5/6.0 (Listening 6.0, Speaking 6.0); CET-6 631